

Good Morning, Prime Minister:

Global Carbon Abatement, Heterogeneous Costs, & the Limits of Voluntary Multilateralism

1. The Scenario

COP30, the most recent iteration of the UNFCCC's annual global climate summit, was held in Belém, Brazil in November 2025. Framed as the 'COP of Implementation' and marking the tenth anniversary of the Paris Agreement, the summit stood as a critical juncture for international climate governance. In an encouraging development, 119 countries submitted updated Nationally Determined Contributions (NDCs) covering approximately 74% of global emissions. Yet independent analysis by the Climate Action Tracker found these plans would still only deliver 15% of the emissions reductions required to remain within 1.5°C of warming (UNFCCC, 2025).

The UK used the summit to cement its status as a self-declared 'Clean Energy Superpower,' submitting an NDC committing to an 81% reduction in greenhouse gas emissions by 2035 relative to 1990 levels. This positioned it as one of the few nations whose plans aligned with the 1.5°C pathway set by the UK's flagship Climate Change Committee (CCC, 2025). Despite this, the Climate Action Network - a global coalition of 1,500+ civil society and NGO members - awarded the UK its informal "Fossil of the Day" award; a title awarded to the country believed to have done the most to obstruct, weaken, or undermine progress on climate action.

This was reflective of the broader disputes between developed and developing countries at COP30, centred around the design of international climate finance. The principle of Common But Differentiated Responsibilities and Respective Capabilities (CBDR-RC), first cited in 1992, states that developed nations bear a greater financial obligation to address climate change by virtue of their historical emissions and superior capabilities (UNFCCC, 1992). At Belém, developing country blocs invoked this principle to argue for differentiated per-unit financial support directed preferentially toward high-cost, lower-income nations as a matter of equity (Heinrich Böll Stiftung, 2025; Mutirão Decision, UNFCCC, 2025). The NCQG's \$1.3 trillion annual target and the Loss and Damage Fund's \$250 million initial disbursement both reflect this equity-driven logic, with developing countries explicitly framing their demands under Article 9.1 of the Paris Agreement as an obligation of developed nations (ODI, 2025).

This essay argues that the CBDR-RC principle, whatever its moral and political virtues, rests on a flawed efficiency argument, and that a well-specified game-theoretic model identifies precisely where and why it fails. The atmosphere is a global public good: non-excludable and non-rival, meaning every tonne of CO₂ pollution avoided generates benefits dispersed across all nations regardless of who bore the abatement cost (Samuelson, 1954). This single property transforms the strategic structure of the problem, wherein the 85% gap between COP30 pledges and the carbon reductions needed to hit the 1.5°C goal is quite simply the Nash Equilibrium. This distinction has profound policy implications, because it shifts the analytical question from how to persuade countries to contribute more and instead focuses on how to restructure payoffs such that greater contribution becomes individually rational.

What makes this richer than a standard Prisoner's Dilemma is that countries are not symmetric. The marginal cost of abatement differs substantially across nations as a function of energy infrastructure, development stage, and industrial composition. For a high-income country with mature renewable capacity such as the UK, the marginal cost of the next unit of abatement is considerably lower than for a coal-dependent emerging economy where fossil fuels remain the primary path to electrification (IPCC, 2022). The model will show, however, that this cost heterogeneity does not generate the policy conclusion that CBDR-RC proponents assume: the efficient Pigouvian correction requires a uniform per-unit subsidy rate across all countries, regardless of cost structure. The correct case for directing larger finance flows toward developing nations rests not on equity-based rate differentiation, but on reducing their marginal abatement costs through technology transfer and green infrastructure, a distinct and more defensible claim. This essay sets out to: (i) model the global abatement problem as a continuous Public Goods Game with country-specific cost parameters; (ii) derive the asymmetric Nash Equilibrium and social optimum, showing that the efficient correction requires a uniform Pigouvian subsidy rate regardless of cost heterogeneity; and (iii) apply these results to evaluate the CBDR-RC equity claims that dominated COP30's finance negotiations.

2. The Formal Model

Players: A set $N = \{1, \dots, N\}$ of sovereign states, each a unitary rational actor maximising national welfare.

Strategies: Each country i simultaneously and independently chooses a continuous abatement level $a_i \in [0, a_{max}]$, interpreted as a range from no abatement effort to the maximum effort scientifically/economically possible. This abatement effort can be proxied/measured either by the fraction of GDP invested in emissions reduction, or by the absolute quantity abated (GtCO₂). We can treat the two as proportional under the assumption of a country-specific but locally constant baseline marginal abatement cost (Nordhaus, 2017).

Let $A = \sum_j a_j$ denote aggregate global abatement and $A_{-i} = A - a_i$ the abatement of all other parties.

Payoffs: Country i 's payoff is:

$$\pi_i(a_i, A_{-i}) = \alpha \cdot A - (c_i/2) \cdot a_i^2 \rightarrow \text{substituting } A = a_i + A_{-i} \rightarrow \pi_i(a_i, A_{-i}) = \alpha(a_i + A_{-i}) - (c_i/2)a_i^2$$

The parameter $\alpha > 0$ is the symmetric per-country marginal benefit of global abatement, encompassing avoided climate damages, health co-benefits, and energy security gains documented in IPCC AR6 (2022); symmetry reflects the non-rival, non-excludable nature of the climate. The country-specific parameter $c_i > 0$ is i 's marginal cost coefficient, capturing heterogeneity in abatement technology, energy infrastructure, and development stage. The convex cost term $(c_i/2)a_i^2$ reflects the empirically well-documented upward-sloping marginal abatement cost curve (Kesicki and Strachan, 2011): early interventions (onshore wind, energy efficiency) are cheap, while later units (green hydrogen, direct air capture) are expensive. Critically, because A_{-i} enters i 's payoff with the same coefficient α as a_i itself, any country's abatement is a perfect substitute for any others', which is the formal expression of the free-rider incentive.

3. Solution

Nash Equilibrium. Country i maximises π_i over a_i taking A_{-i} as given. Differentiating:

$$\partial \pi_i / \partial a_i = \alpha - c_i \cdot a_i = 0 \implies a_i^* = \alpha / c_i$$

The second-order condition $\partial^2 \pi_i / \partial a_i^2 = -c_i < 0$ confirms a maximum. Crucially, the best response is independent of A_{-i} : abating at α/c_i is a dominant strategy. Whatever other nations pledge at Belém, country i 's individually rational response is uniquely determined by its own cost parameter alone (no beliefs about others are required). The unique asymmetric Nash Equilibrium is therefore the vector:

$$a^{NE} = (\alpha/c_1, \alpha/c_2, \dots, \alpha/c_n)$$

With higher-cost countries (large c_i) abating *less* in equilibrium. The UK, with relatively mature renewable infrastructure and an existing ETS generating £4.8bn annually (DESNZ, 2024), operates with a comparatively low c_{UK} , implying higher equilibrium abatement than coal-dependent emerging economies. This heterogeneity is important in the welfare analysis that follows.

Social Optimum. A global social planner maximises $W = \sum_i \pi_i = N\alpha \cdot A - \sum_i (c_i/2) \cdot a_i^2$. Differentiating with respect to each a_i :

$$\partial W / \partial a_i = N \cdot \alpha - c_i \cdot a_i = 0 \implies a_i^{SO} = N\alpha / c_i$$

The planner internalises the positive externality that its abatement confers on all N countries, hence the N -fold scaling of the marginal benefit. The efficiency ratio for every country i is:

$$a_i^* / a_i^{SO} = (\alpha / c_i) / (N\alpha / c_i) = 1/N$$

The $1/N$ efficiency ratio holds for every country regardless of its cost parameter c_i . At $N = 119$ COP30 parties, $1/N = 0.0084$, meaning every signatory abates at just 0.84% of the socially optimal level. Notably, the social optimum allocates abatement inversely proportional to cost, meaning low- c_i countries (developed) are directed to abate disproportionately more than high-cost (developing) countries.

Pigouvian Correction. If each country receives a per-unit abatement subsidy s (financed via an international climate fund), the modified best response becomes $a_i^*(s) = (\alpha + s) / c_i$. Setting this equal to $a_i^{SO} = N\alpha / c_i$ and solving:

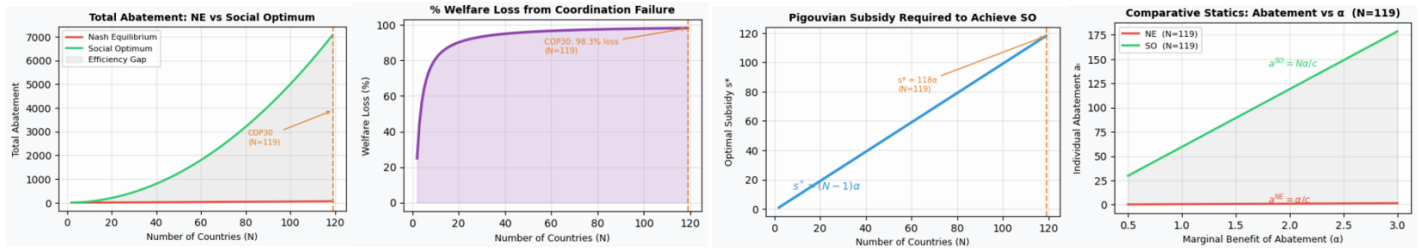
$$s^* = (N - 1) \cdot \alpha$$

The optimal subsidy is independent of c_i , meaning every country requires the same per-unit correction. This is because the positive externality that any country's abatement gives to other countries is α , which is symmetric across all players regardless of their cost parameter. The per-unit subsidy required to internalise that foregone benefit is identical for a coal-dependent emerging economy and a mature renewable-energy nation alike. For $N = 119$, $s^* = 118\alpha$, an 11,800% uplift on the unilateral incentive, highlighting precisely why voluntary pledges are structurally inadequate.

4. Simulation

While the Nash Equilibrium, social optimum, and Pigouvian subsidy are each derivable analytically in closed form, a Python simulation can be used to make the scale and trajectory of the inefficiency empirically transparent. Plotting welfare loss as a continuous curve from $N = 2$ to $N = 119$, rising steeply before flattening at 98.33% at the COP30 marker, communicates the severity of the coordination failure more immediately than the expression $(W^{SO} - W^{NE}) / W^{SO}$ alone, and allows direct comparison against the observed 85% ambition gap at Belém. The simulation serves three purposes: confirming analytical results hold across the full empirically relevant range of N ; calibrating the theoretical efficiency ratio $1/N$ to the specific context of COP30; and translating the abstract Pigouvian subsidy $s^* = (N-1)\alpha$ into concrete dollar figures by setting $\alpha = \text{SCC}/N$ (each country's equal share of the global benefit of one tonne abated). This yields $s^* \approx 99.2\%$ of the full Social Cost of Carbon at $N = 119$, and a total corrective transfer of approximately \$5.1 trillion annually against the NCQG's \$1.3 trillion target, using Rennert et al.'s (2022) peer-reviewed estimate of \$185/tonne. This figure reveals the true scale of the gap between current multilateral commitments and the welfare-maximising solution. The full simulation code is available at: github.com/morgansmith323/climate-public-goods-game

Global Carbon Abatement: Public Goods Game Simulation
 $n_i = \alpha \sum a_j - \frac{1}{2} a_i^2 \mid \alpha = 1, c = 2 \mid N = 2 \text{ to } 119 \text{ (COP30)}$



5. Conclusion

The Public Goods Game with heterogeneous costs delivers a verdict on the COP30 finance debate that is both precise and uncomfortable. The Nash Equilibrium is unique, asymmetric, and Pareto-dominated: every country under-provides abatement by a factor of N regardless of its cost structure, producing a welfare loss of 98.33% at $N = 119$. As shown above, every signatory abates at just 0.84% of the socially optimal level, therefore implying a theoretical shortfall of 99.16% (even worse than the observed 85% ambition gap at Belém). The real-world gap is somewhat smaller than the pure Nash prediction, likely due to the partial internalisation of domestic co-benefits such as air quality and energy security (which effectively raises each country's private α above the baseline). There are also reputational pressures, and conditional pledge structures that introduce weak strategic complementarities absent in the static model above.

The uniform Pigouvian correction $s^* = (N-1)\alpha$ (simulated at ~\$5.1 trillion annually, four times the NCQG's \$1.3 target) delivers the essay's most consequential contribution to the climate finance debate. The CBDR-RC principle states that high-cost, lower-income nations are entitled to preferentially larger per-unit financial transfers; but the model argues the opposite. Since the externality being corrected is symmetric, meaning every country's abatement confers the same marginal benefit α on every other party regardless of cost structure, the efficient per-unit subsidy rate is identical across all 119 signatories. This undermines the efficiency argument underlying the CBDR-RC call for rate differentiation. The correct first-principles case for directing larger finance flows toward developing nations is the reduction of c_i itself: funding the advancement of technology and green infrastructure in high-cost, developing nations will lower marginal abatement costs, in turn shifting their contribution toward the social optimum level over time. Measured against this standard, the NCQG is best understood not as an equity instrument calibrated to historical emissions obligations, but as a cost-reduction mechanism – which, at \$1.3 trillion, remains structurally and substantially below the scale that the model demands.

The model's most important limitation, however, points toward an even more insightful strategic structure. The current specification assumes no strategic complementarities, meaning one country's abatement neither raises nor lowers another's incentive to abate. When we introduce a spillover parameter $\beta > 0$ capturing technology diffusion effects (e.g. UK offshore wind investment demonstrably reduced global turbine costs; TBI, 2022; IRENA, 2024), the best response functions become upward-sloping in other players' abatement. Above a critical threshold level of β^* , the game transitions from a Public Goods Game with a unique 'bad' equilibrium into a Stag Hunt with two equilibria: a low-abatement trap and a high-abatement cooperative equilibrium (Skyrms, 2004). Crucially, this shifts the policy lever from price correction (Pigouvian Subsidy) toward expectation coordination: credible technology leadership lowers global c_i and in turn changes what countries expect of each other, making the high-abatement equilibrium the natural point of convergence. If the UK's Clean Energy Superpower agenda results in significant reductions in global c_i through technology export, it may do more to resolve the international coordination failure than any financial pledge calibrated to CBDR-RC obligations. The game theory ultimately suggests that the most powerful contribution the UK can make at COP30 (and beyond) is a technology strategy that makes ambitious abatement the individually rational choice for every player at the table.

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